

NOVARA CAMERI, Italy

WMO: 160640

Lat: 45.530N

Lon: 8.667E

Elev: 586

StdP: 14.39

Time Zone: 1.00 (EUC)

Period: 90-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	21.5	-0.4	5.5	49.7	7.2	8.2	46.3	22.1	43.9	18.1	45.4	1.8	10	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	88.0	71.1	86.1	70.7	84.2	69.3	75.3	84.3	73.6	82.4	72.0	79.8	4.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.0	121.0	79.5	71.2	117.6	78.5	69.5	111.0	76.4	39.1	84.1	37.5	82.0	36.0	81.4	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.1	13.6	10.9	DB	13.3	91.0	2.5	1.7	11.6	92.2	10.1	93.1	8.8	94.1	7.0	95.3	
			WB	13.2	77.0	2.3	2.6	11.5	78.9	10.2	80.4	8.9	81.8	7.2	83.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.4	35.8	39.7	47.8	52.7	62.7	68.3	72.7	72.9	64.8	54.5	43.8	36.3	(6)
	DBStd	14.16	4.58	5.70	5.26	5.04	4.98	5.15	4.13	4.10	4.48	5.76	5.67	5.35	(7)
	HDD50	1511	440	290	102	30	1	0	0	0	0	25	198	425	(8)
	HDD65	4557	905	709	532	370	101	25	1	3	58	326	637	890	(9)
	CDD50	3123	0	1	35	110	395	549	704	709	444	165	11	0	(10)
	CDD65	694	0	0	0	0	30	124	240	247	52	1	0	0	(11)
	CDH74	5936	0	0	3	8	282	1085	2175	2040	335	8	0	0	(12)
	CDH80	1671	0	0	0	0	34	255	691	657	33	1	0	0	(13)
(14) Wind	WSAvg	3.6	2.8	3.5	4.6	4.9	4.7	4.2	3.5	3.4	3.1	3.2	3.1	2.6	(14)
(15) Precipitation	PrecAvg	42.8	2.8	3.2	3.9	3.2	4.8	3.7	2.6	3.9	3.5	4.7	4.8	2.2	(15)
	PrecMax	66.5	8.1	10.6	11.0	8.9	13.1	7.6	6.3	9.4	10.4	13.3	8.9	5.1	(16)
	PrecMin	28.2	0.1	0.0	0.0	0.0	0.9	1.1	0.6	0.3	0.0	0.0	0.3	0.2	(17)
	PrecStd	11.7	2.0	2.6	2.9	2.2	3.3	2.0	1.5	2.3	2.9	4.0	2.7	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	64.3	72.0	75.1	83.9	89.8	89.9	90.0	83.8	75.0	62.5	55.1	(19)
		MCWB	42.0	48.6	53.6	57.0	64.6	70.5	72.0	71.3	66.6	63.6	50.0	41.8	(20)
	2%	DB	50.1	58.7	67.6	71.2	80.3	85.8	88.0	88.0	80.6	70.1	58.7	49.9	(21)
		MCWB	41.5	46.4	52.3	54.2	63.2	69.2	71.1	71.5	64.7	59.6	50.0	41.9	(22)
	5%	DB	46.7	55.3	64.2	68.0	77.0	82.8	86.1	86.1	78.4	67.2	55.5	46.8	(23)
		MCWB	40.6	44.3	50.7	53.5	61.8	68.1	70.9	71.4	64.0	58.0	50.5	43.2	(24)
	10%	DB	44.4	51.6	60.7	64.7	74.4	80.4	84.2	84.1	75.3	64.3	53.4	45.2	(25)
		MCWB	40.4	43.2	49.2	51.4	61.3	66.6	69.4	70.1	63.1	57.8	49.6	42.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.4	51.4	57.3	59.9	68.0	74.8	78.5	76.7	70.0	64.9	55.7	48.6	(27)
		MCDB	49.3	59.0	67.4	69.0	78.5	82.9	86.4	85.4	77.6	70.5	58.5	49.5	(28)
	2%	WB	44.0	48.4	54.8	57.4	65.7	72.7	75.9	74.9	68.2	62.9	53.7	46.4	(29)
		MCDB	46.3	56.2	63.3	65.5	75.8	81.9	84.9	84.5	75.1	67.3	55.9	47.2	(30)
	5%	WB	42.2	46.1	52.8	55.6	63.9	70.5	74.2	73.2	67.0	61.1	52.2	44.7	(31)
		MCDB	44.3	51.7	61.0	63.4	72.7	79.0	82.6	82.1	73.9	64.8	54.1	45.6	(32)
	10%	WB	40.5	44.1	50.7	54.1	62.5	68.7	72.4	71.8	65.5	59.5	50.6	42.6	(33)
		MCDB	42.5	48.6	57.3	61.8	71.6	77.3	79.8	79.6	72.0	62.9	52.4	44.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	19.2	21.3	19.3	18.8	19.4	20.1	19.6	18.7	15.8	15.0	14.6	(35)
		MCDBR	21.0	27.2	28.7	25.7	23.5	22.0	22.1	22.1	22.8	20.8	19.7	18.4	(36)
	5% WB	MCWBR	14.1	16.3	15.6	12.2	10.4	9.9	10.1	9.8	9.8	11.4	12.6	12.4	(37)
		MCWBR	15.4	22.6	25.2	20.2	20.9	20.5	20.4	20.9	17.7	16.7	12.4	11.5	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.343	0.372	0.422	0.468	0.469	0.489	0.470	0.464	0.452	0.434	0.376	0.336	(40)
		taud	2.355	2.259	2.138	2.049	2.104	2.091	2.146	2.166	2.157	2.162	2.288	2.381	(41)
	Ebn at Noon	Ebn at Noon	237	251	252	251	256	250	254	250	240	223	219	226	(42)
		Edh at Noon	25	32	42	50	49	50	47	44	41	36	26	22	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	460	712	1135	1428	1758	2028	2065	1747	1310	770	455	383	(44)
	RadStd	RadStd	66	103	144	162	143	119	128	108	90	120	81	55	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (76 neighbors)	+0.58	N/A	N/A	+1.22	+0.72	+0.56	N/A	N/A	+103	+18

Nomenclature: See separate page